
Unmixing the Music: Sparse, Low-Rank, and Deep Compressed-Sensing Approaches to Music Source Separation

Taka Khoo¹

Abstract

Music source separation, the recovery of individual stems such as vocals and accompaniment from a single mixture, is usually approached either by classical signal-processing decompositions or by large end-to-end neural networks. We show that a wide span of these methods are instances of one underlying principle: recovery of structured (sparse or low-rank) representations of the time-frequency spectrogram. Under this view we construct and evaluate four separators on a common short-time Fourier transform front end: a low-rank-plus-sparse decomposition by principal component pursuit, a supervised non-negative matrix factorization with per-source dictionaries, a discriminative K-SVD dictionary model with sparse-representation classification, and a deep sparse coding network that predicts a time-frequency mask by unrolling a non-negative proximal-gradient solver and learning its dictionaries end-to-end. We further give a joint-sparsity extension for stereo recovery. On 25 held-out tracks of the MUSDB18 benchmark, the methods form a clear ordering: training-free baselines (median-filtering harmonic percussive separation, REPET-SIM) trail supervised dictionary methods, and the unrolled deep sparse coding network is the strongest non-oracle separator on both sources (vocal SI-SDR 2.0 dB, accompaniment SI-SDR 8.3 dB) while being the fastest learned model at inference. We connect each method to the recovery theory that governs it and report a negative result: the learned dictionaries are highly coherent ($\mu \approx 0.85$), so worst-

case exact-recovery guarantees are vacuous at the operating sparsity, yet separation succeeds, an instance of the average-case-versus-worst-case gap for trained dictionaries. Code, trained models, and audio are released.

1. Introduction

Single-channel music source separation asks for the constituent stems of a recording given only the final mixture. It underpins remixing, upmixing, music education, and downstream music-information-retrieval tasks such as transcription and lyric alignment, and it remains hard because the mixing process is information-destroying: many stem configurations explain the same mixture (Cano et al., 2019; Araki et al., 2025). Progress has come from two largely separate traditions. The first is model-based: non-negative matrix factorization (NMF) (Lee & Seung, 1999; Smaragdis, 2007), low-rank-plus-sparse decompositions (Huang et al., 2012), and dictionary learning (Aharon et al., 2006) exploit explicit structural priors on the spectrogram. The second is data-driven: deep networks trained end-to-end, including U-Nets (Jansson et al., 2017; Stoller et al., 2018), recurrent masking models (Stöter et al., 2019), and hybrid waveform/spectrogram architectures (Défossez et al., 2021; Rouard et al., 2023), define the current state of the art.

This paper takes the position that these traditions are two ends of a single axis. The spectrogram magnitude of a musical source is approximately sparse in a source-adapted basis (a vocal frame activates a few formant atoms; a percussive frame is sparse in time) and the sustained harmonic content of an accompaniment is approximately low-rank. These two structures are not separate assumptions but two sides of the same coin: the rank of a matrix is the ℓ_0 norm of its singular spectrum, and the nuclear norm is the corresponding ℓ_1 relaxation, so “few active atoms” and “few active singular vectors” are the same prior applied to a vector and to a matrix. Both are exactly the structural assumptions of compressed sensing and

¹Thayer School of Engineering, Dartmouth College, Hanover, NH, USA. Correspondence to: Taka Khoo <taka.khoo.th@dartmouth.edu>.

sparse coding (Donoho, 2006; Elad, 2010). The recovery problem is in every case the underdetermined system $\mathbf{y} = \mathbf{A}\mathbf{x}$ with a sparse code: the classical half of the subject fixes the dictionary $\mathbf{A}$ and recovers the code, while dictionary learning, K-SVD, and NMF instead learn $\mathbf{A}$ from data. Deep masking networks sit at the learned end of this same continuum: a sparse coding network is, in a precise sense, a learned autoencoder (Gregor & LeCun, 2010; Pappayan et al., 2017), an unrolled sparse-recovery solver whose operators are trained by backpropagation (Monga et al., 2021). Placing the classical and deep methods on one short-time Fourier transform (STFT) front end therefore lets us compare them as points on a single structure-recovery spectrum rather than as unrelated systems.

Contributions. (i) We formalize vocal/accompaniment separation as a structured-recovery problem and instantiate four separators that share a front end and an evaluation harness (Section 5). (ii) We construct a deep sparse coding network that predicts a soft time-frequency mask by unrolling a non-negative fast proximal-gradient iteration and learning the synthesis dictionaries end-to-end, and show it is the strongest non-oracle separator in our study while being the fastest learned model at test time (Sections 5.4 and 6). (iii) We pair every method with the recovery condition that governs it, and report an honest negative result on dictionary coherence that quantifies the gap between worst-case guarantees and observed behaviour (Sections 4 and 6.1). All code, trained weights, and audio examples are released for reproducibility.

2. Background: Spectrograms, Stems, and Masking

We summarize the audio concepts the paper relies on so the development is self-contained for readers outside music processing.

From waveform to spectrogram. A recording is a pressure waveform $y(t)$ sampled at f_s samples per second. The short-time Fourier transform (STFT) slides a window of length w across the signal with hop h and takes a discrete Fourier transform in each window, producing a complex matrix $\mathbf{Y} \in \mathbb{C}^{F \times T}$ where $F = w/2 + 1$ frequency bins index pitch and T frames index time. Its modulus $\mathbf{M} = |\mathbf{Y}|$, the magnitude spectrogram, is a nonnegative image whose brightness at (f, t) is the energy near frequency f at time t ; its argument $\angle \mathbf{Y}$ is the phase. The transform is invertible, so y can be reconstructed exactly from $\mathbf{M}$ and $\angle \mathbf{Y}$.

Stems and the separation goal. A finished song is a sum of source stems (vocals, drums, bass, other). We target the two-way split into vocals and accompaniment (everything else). Because the STFT is linear, the mixture spectrogram is approximately the sum of the source spectrograms, $\mathbf{Y} \approx \sum_c \mathbf{Y}_c$, and on magnitudes the sources mostly occupy different time-frequency cells, which is what makes masking viable.

Time-frequency masking. Rather than predict a source spectrogram outright, we predict a mask $\mathbf{M}_c \in [0, 1]^{F \times T}$ that scales each cell of the mixture and resynthesizes with the mixture phase,

$$\hat{y}_c = \text{STFT}^{-1}(\mathbf{M}_c \odot \mathbf{Y}). \quad (1)$$

A mask answers a local question: what fraction of the energy in cell (f, t) belongs to source c ? The optimal mask under a squared-error criterion is the ideal ratio mask

$$\mathbf{M}_c^{\text{IRM}} = \frac{|\mathbf{Y}_c|^2}{\sum_k |\mathbf{Y}_k|^2}, \quad (2)$$

which is exactly the multichannel Wiener / minimum-mean-square-error estimator when sources are modelled as locally uncorrelated Gaussians. Equation (2) requires the true sources, so it serves only as an oracle upper bound; the binary variant $\mathbf{M}_c^{\text{IBM}} = \mathbb{I}[|\mathbf{Y}_c| > |\mathbf{Y}_{-c}|]$ is a second oracle. Every method we build estimates a mask without access to the sources, and no mixture-phase magnitude mask can exceed the ideal-mask ceilings.

Why sparsity and low rank. Two empirical regularities make these masks recoverable by structured optimization. A voiced sound places energy on a sparse set of harmonics (a few bright horizontal ridges per frame), so its spectrogram is approximately sparse in a harmonic-atom basis. Sustained, repeating accompaniment reuses a small set of spectral templates across many frames, so its magnitude spectrogram is approximately low-rank. These are the two structural priors compressed sensing is built to exploit, and they organize the four methods below.

3. Related Work

Factorization and low-rank models. NMF decomposes a non-negative spectrogram into non-negative spectral atoms and activations (Lee & Seung, 1999; 2000) and was applied to polyphonic audio by Smaragdis & Brown (2003). Supervised and convolutive variants (Smaragdis, 2007; Virtanen, 2007) pre-train one dictionary per source and hold it fixed at test time, which is the form we adopt. The choice

of divergence matters for audio: the Itakura-Saito divergence (Févotte et al., 2009) is scale-invariant and matches a Gaussian generative model of power spectra. Low-rank methods exploit the repetitive structure of accompaniment: REPET (Rafi & Pardo, 2013) models the background as a repeating pattern, and robust principal component analysis (RPCA) separates a low-rank accompaniment from a sparse vocal (Candès et al., 2011; Huang et al., 2012). Median-filtering harmonic/percussive separation (FitzGerald, 2010) is a training-free baseline in the same family.

Dictionary learning and sparse representation. K-SVD learns overcomplete sparsifying dictionaries by alternating sparse coding with rank-one atom updates (Aharon et al., 2006; Engan et al., 1999). Sparse-representation classification (SRC) assigns a signal to the class whose sub-dictionary best reconstructs it (Wright et al., 2009); we use this rule to route time-frequency energy. Task-driven dictionary learning (Mairal et al., 2012) optimizes the dictionary for a downstream loss by differentiating through the sparse code, which is the conceptual bridge to our learned model.

Deep separation and algorithm unrolling. End-to-end networks dominate the leaderboards (Hennequin et al., 2020; Stöter et al., 2019; Défossez et al., 2021; Rouard et al., 2023; Luo & Mesgarani, 2019; Mitsufuji et al., 2022). A complementary line interprets such networks through sparse coding: LISTA unrolls the iterative shrinkage-thresholding algorithm into a fixed-depth trainable network (Gregor & LeCun, 2010); CNNs admit a multilayer convolutional-sparse-coding reading (Papayan et al., 2017); and algorithm unrolling is now a general design principle (Monga et al., 2021; Sprechmann et al., 2015). Discriminatively unfolded NMF has been used for speech separation (Le Roux et al., 2015), and supervised deep sparse coding networks have been studied for image classification (Sun et al., 2019). Our learned separator instantiates this unrolling idea for music spectrogram masking.

4. Preliminaries: Structured Recovery

We collect the recovery results that the methods rely on. Throughout, $\|\mathbf{x}\|_0 = |\text{supp}(\mathbf{x})|$ counts nonzeros and $\|\mathbf{x}\|_1 = \sum_i |x_i|$.

Sparse recovery. Given $\mathbf{A} \in \mathbb{R}^{m \times N}$ with $m \ll N$ and $\mathbf{y} = \mathbf{A}\mathbf{x}$ for an s -sparse $\mathbf{x}$, the combinatorial program $\min_{\mathbf{z}} \|\mathbf{z}\|_0$ s.t. $\mathbf{A}\mathbf{z} = \mathbf{y}$ is NP-hard; its convex relaxation $\min_{\mathbf{z}} \|\mathbf{z}\|_1$ s.t. $\mathbf{A}\mathbf{z} = \mathbf{y}$ is a linear program. Exactness of the relaxation is controlled by three quan-

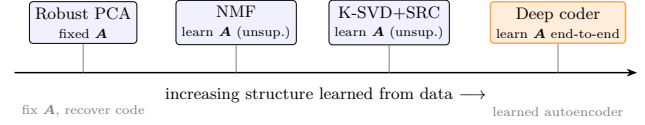


Figure 1. The four separators as points on one structure-recovery spectrum. All solve the same underdetermined system $\mathbf{y} = \mathbf{A}\mathbf{x}$ with a sparse code; they differ only in how much of the dictionary $\mathbf{A}$ is learned from data, from a fixed convex split to a fully trained unrolled solver.

ties. The spark is the size of the smallest linearly dependent column subset, and an s -sparse solution is unique iff $\text{spark}(\mathbf{A}) > 2s$. The restricted isometry property (RIP) of order s holds with constant δ_s if

$$(1 - \delta_s)\|\mathbf{x}\|_2^2 \leq \|\mathbf{A}\mathbf{x}\|_2^2 \leq (1 + \delta_s)\|\mathbf{x}\|_2^2 \quad (3)$$

for all s -sparse $\mathbf{x}$; ℓ_1 recovery is exact whenever $\delta_{2s} < \sqrt{2} - 1$ (Candès & Tao, 2005). The mutual coherence $\mu(\mathbf{A}) = \max_{i \neq j} |\langle \mathbf{A}_i, \mathbf{A}_j \rangle|$ (unit-norm columns) gives the verifiable condition

$$s < \frac{1}{2} \left(1 + \frac{1}{\mu(\mathbf{A})} \right) \quad (4)$$

under which both orthogonal matching pursuit (OMP) and ℓ_1 minimization recover every s -sparse signal (Tropp & Gilbert, 2007; Donoho, 2006). For sub-Gaussian random $\mathbf{A}$, RIP holds with high probability once $m \geq C s \log(N/s)$.

Low-rank recovery. The nuclear norm $\|\mathbf{X}\|_* = \sum_i \sigma_i(\mathbf{X})$ is the convex surrogate for rank. Robust PCA recovers a low-rank $\mathbf{X}$ and a sparse $\mathbf{E}$ from $\mathbf{M} = \mathbf{X} + \mathbf{E}$ by principal component pursuit,

$$\min_{\mathbf{X}, \mathbf{E}} \|\mathbf{X}\|_* + \lambda \|\mathbf{E}\|_1 \quad \text{s.t.} \quad \mathbf{X} + \mathbf{E} = \mathbf{M}, \quad (5)$$

which succeeds with high probability at $\lambda = 1/\sqrt{\max(n_1, n_2)}$ when the low-rank term is incoherent and the support of $\mathbf{E}$ is random (Candès et al., 2011; Candès & Recht, 2009).

5. Method

Front end. Let $y(t)$ be a mono mixture. Its STFT is $\mathbf{Y} = \text{STFT}(y) \in \mathbb{C}^{F \times T}$ with magnitude $\mathbf{M} = |\mathbf{Y}|$ and phase $\angle \mathbf{Y}$. Every separator produces a soft mask $\mathbf{M}_c \in [0, 1]^{F \times T}$ per source c , and resynthesizes $\hat{y}_c = \text{STFT}^{-1}(\mathbf{M}_c \odot \mathbf{Y})$ using the mixture phase. Given non-negative source magnitude estimates $\widehat{\mathbf{M}}_c$, we use Wiener-style masks $\mathbf{M}_c = \widehat{\mathbf{M}}_c^2 / \sum_k \widehat{\mathbf{M}}_k^2$. Figure 1 places the four methods of this section on the structure-recovery spectrum the introduction describes.

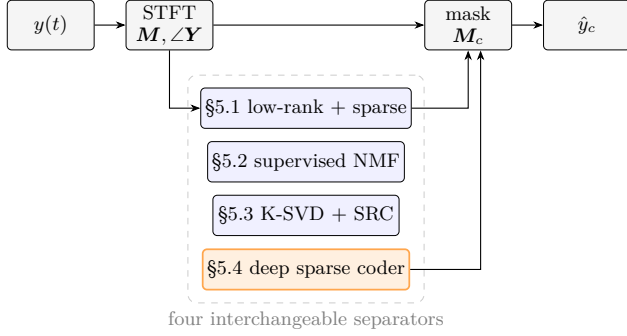


Figure 2. The shared pipeline. All four methods estimate the same time-frequency mask $\mathbf{M}_c$ on one STFT front end and differ only in how they impose structure; resynthesis reuses the mixture phase.

Table 1. The four separators as one family: each pairs a structural prior with a recovery penalty and a learning mode. $\|\cdot\|_*$ is the nuclear norm.

Method	Prior	Penalty	Learning
RPCA	low-rank+sparse	nuclear+ ℓ_1	none
NMF	nonneg. parts	KL div.	unsup.
K-SVD	sparse code	$\ell_0 \leq s$	unsup.
SCN	unrolled sparse	elastic-net	end-to-end

Figure 2 shows the shared scaffolding. The methods differ only in the structural prior they place on the spectrogram, which lets us read them as points on a single structure-recovery axis (Table 1).

5.1. Low-rank plus sparse (RPCA)

We solve Equation (5) on the magnitude spectrogram by the inexact augmented Lagrange multiplier method. With the soft-threshold $\text{soft}_\tau(x) = \text{sgn}(x) \max(|x| - \tau, 0)$ and the singular-value threshold $\text{SVT}_\tau(\mathbf{Z}) = \mathbf{U} \text{soft}_\tau(\Sigma) \mathbf{V}^\top$ for $\mathbf{Z} = \mathbf{U} \Sigma \mathbf{V}^\top$, the iteration is

$$\mathbf{X} \leftarrow \text{SVT}_{1/\mu}(\mathbf{M} - \mathbf{E} + \frac{1}{\mu} \mathbf{A}), \quad (6)$$

$$\mathbf{E} \leftarrow \text{soft}_{\lambda/\mu}(\mathbf{M} - \mathbf{X} + \frac{1}{\mu} \mathbf{A}), \quad (7)$$

$$\mathbf{A} \leftarrow \mathbf{A} + \mu(\mathbf{M} - \mathbf{X} - \mathbf{E}), \quad (8)$$

with μ increased geometrically. The low-rank $\mathbf{X}$ is the accompaniment estimate and the sparse $\mathbf{E}$ the vocal proxy; this is principal component pursuit applied to singing-voice separation (Huang et al., 2012). The intuition is the same one that separates a static background from moving foreground in surveillance video: the repeating accompaniment is the low-rank “background” of the spectrogram, and the voice, like a scratch on a CD or a rare gross corruption, is the sparse part that classical PCA would smear but principal component pursuit isolates. The method is training-

free.

5.2. Supervised NMF

We pre-train one non-negative dictionary per source on isolated stems by minimizing $\|\mathbf{M}_c - \mathbf{D}_c \mathbf{H}_c\|$ subject to $\mathbf{D}_c, \mathbf{H}_c \geq 0$. With the Kullback-Leibler divergence (the natural loss for magnitude spectra (Févotte et al., 2009)), the Lee-Seung multiplicative updates (Lee & Seung, 2000) are

$$\mathbf{H} \leftarrow \mathbf{H} \odot \frac{\mathbf{D}^\top (\mathbf{M} \oslash \mathbf{D} \mathbf{H})}{\mathbf{D}^\top \mathbf{1}}, \quad \mathbf{D} \leftarrow \mathbf{D} \odot \frac{(\mathbf{M} \oslash \mathbf{D} \mathbf{H}) \mathbf{H}^\top}{\mathbf{1} \mathbf{H}^\top}, \quad (9)$$

where $\odot, \oslash$ are elementwise. These updates are not heuristic: they are a majorization-minimization scheme. For the divergence $D(\mathbf{M} \parallel \mathbf{D} \mathbf{H})$ one builds an auxiliary function $G(\mathbf{H}, \mathbf{H}') \geq D(\mathbf{M} \parallel \mathbf{D} \mathbf{H})$ with equality at $\mathbf{H}' = \mathbf{H}$, using the convexity of $-\log$ and Jensen’s inequality on the latent assignment $\mathbf{D}_{ik} \mathbf{H}_{kj} / (\mathbf{D} \mathbf{H})_{ij}$; minimizing $G(\cdot, \mathbf{H}')$ in closed form yields the multiplicative step, so each update cannot increase the loss and the iterates converge to a stationary point. The factors stay nonnegative automatically because every quantity in the ratio is nonnegative. At test time we fix $\mathbf{D} = [\mathbf{D}_v \mid \mathbf{D}_a]$ and solve only for activations $\mathbf{H}$, then reconstruct $\hat{\mathbf{M}}_v = \mathbf{D}_v \mathbf{H}_v$ and $\hat{\mathbf{M}}_a = \mathbf{D}_a \mathbf{H}_a$. This is the supervised NMF template of Smaragdis (2007); the Kullback-Leibler choice corresponds to a Poisson observation model on spectrogram counts, while the Itakura-Saito divergence ($\beta=0$) corresponds to a multiplicative-Gaussian model of power spectra (Févotte et al., 2009).

5.3. Discriminative dictionaries with SRC

We replace each NMF dictionary with an overcomplete K-SVD dictionary trained to sparsely code spectrogram frames, $\min_{\mathbf{D}, \mathbf{X}} \|\mathbf{Y} - \mathbf{D} \mathbf{X}\|_F^2$ s.t. $\|\mathbf{x}_i\|_0 \leq s$. K-SVD (Aharon et al., 2006) alternates OMP sparse coding with a per-atom update: writing $\mathbf{D} \mathbf{X} = \sum_j \mathbf{D}_j \mathbf{x}_j^\top$ and forming the residual $\mathbf{E}_j = \mathbf{Y} - \sum_{k \neq j} \mathbf{D}_k \mathbf{x}_k^\top$ restricted to the atom’s active set, the rank-one SVD $\mathbf{E}_j^R = \mathbf{U} \Sigma \mathbf{V}^\top$ updates $\mathbf{D}_j \leftarrow \mathbf{u}_1$ and the coefficients $\leftarrow \sigma_1 \mathbf{v}_1$. The rank-one update is optimal in the Eckart-Young sense: $(\mathbf{u}_1, \sigma_1 \mathbf{v}_1)$ is the exact minimizer of $\|\mathbf{E}_j^R - \mathbf{D}_j \mathbf{x}_j^\top\|_F$ over all unit atoms and coefficient rows, so the atom update is a coordinate descent step that cannot increase the objective, and restricting to the active set $\{i : \mathbf{x}_i^j \neq 0\}$ preserves the sparsity pattern set by the coding step. At test time we sparse-code each mixture frame on $[\mathbf{D}_v \mid \mathbf{D}_a]$ by OMP and route energy by the SRC residual rule (Wright et al., 2009): $\mathbf{c}^* = \arg \min_{\mathbf{c}} \|\mathbf{y} - \mathbf{D}_c \mathbf{c}\|_2$. The coherence condition Equation (4) governs when this per-

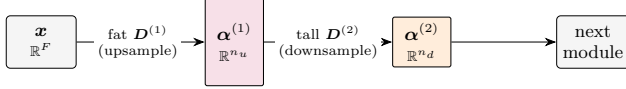


Figure 3. One composite module. A fat dictionary lifts the frame to a tall over-complete sparse code $\alpha^{(1)}$; a tall dictionary compresses it to a short code $\alpha^{(2)}$. Stacking L_m of these and a sigmoid head emits the mask. The expand-then-reduce shape is the autoencoder hourglass with a sparse bottleneck.

frame code is unique, linking the separator’s reliability to a measurable property of the learned dictionary (Section 6.1).

5.4. Deep sparse coding network

Our learned separator predicts the vocal mask by unrolling a non-negative sparse solver and learning its dictionaries end-to-end. The architecture makes the “learned autoencoder” reading of sparse coding literal: each module first lifts a frame into an over-complete sparse code with a fat dictionary, then compresses that code with a tall dictionary, an expand-then-reduce hourglass whose bottleneck is a sparse representation (Figure 3). For an input frame $\mathbf{x} \in \mathbb{R}^F$ a composite module computes two stacked non-negative elastic-net codes,

$$\alpha^{(1)} = \arg \min_{\alpha \geq 0} \frac{1}{2} \|\mathbf{x} - \mathbf{D}^{(1)}\alpha\|_2^2 + \lambda_1 \|\alpha\|_1 + \frac{\lambda_2}{2} \|\alpha\|_2^2, \quad (10)$$

$$\alpha^{(2)} = \arg \min_{\alpha \geq 0} \frac{1}{2} \|\alpha^{(1)} - \mathbf{D}^{(2)}\alpha\|_2^2 + \lambda_1 \|\alpha\|_1 + \frac{\lambda_2}{2} \|\alpha\|_2^2, \quad (11)$$

with $\mathbf{D}^{(1)} \in \mathbb{R}^{F \times n_u}$ overcomplete ($n_u > F$) and $\mathbf{D}^{(2)} \in \mathbb{R}^{n_u \times n_d}$ undercomplete ($n_d < n_u$). Each sub-problem is solved by K steps of a non-negative fast proximal-gradient iteration: with step size $1/L$, L the squared spectral norm of the dictionary (obtained by power iteration, so no on-device SVD is needed),

$$\alpha_k = \left[\mathbf{z}_k - \frac{1}{L} \mathbf{D}^\top (\mathbf{D} \mathbf{z}_k - \mathbf{x}) - \frac{\lambda_1}{L} \right]_+, \quad (12)$$

followed by the Nesterov extrapolation $t_{k+1} = \frac{1}{2}(1 + \sqrt{1 + 4t_k^2})$, $\mathbf{z}_{k+1} = \alpha_k + \frac{t_k - 1}{t_{k+1}}(\alpha_k - \alpha_{k-1})$. Every operation is a matrix multiply or a clamp, so the unrolled solver is differentiable and runs on a GPU. Stacking L_m composite modules and a linear head with a sigmoid yields a mask $\mathbf{M}_v \in [0, 1]^{F \times T}$. To give each prediction local time structure, the input is not a single frame but a stacked $\pm c$ -frame context window $\mathbf{x} = [\mathbf{M}_{:,t-c}; \dots; \mathbf{M}_{:,t+c}] \in \mathbb{R}^{(2c+1)F}$, while the target

Algorithm 1 Composite module forward pass (one of L_m). All steps are matrix multiplies and clamps, so the pass is differentiable and runs on a GPU.

-
- 1: Input: frame $\mathbf{x}$, dictionaries $\mathbf{D}^{(1)}, \mathbf{D}^{(2)}$, shrinkage λ_1, λ_2 , steps K
 - 2: $\mathbf{D} \leftarrow \mathbf{D}^{(1)}$; $L \leftarrow \sigma_{\max}(\mathbf{D})^2 + \lambda_2$ (power iteration)
 - 3: $\alpha \leftarrow \mathbf{0}$; $\mathbf{z} \leftarrow \mathbf{0}$; $t \leftarrow 1$
 - 4: for $k = 1$ to K do
 - 5: $\mathbf{g} \leftarrow \mathbf{D}^\top (\mathbf{D} \mathbf{z} - \mathbf{x}) + \lambda_2 \mathbf{z}$ (gradient)
 - 6: $\alpha^+ \leftarrow [\mathbf{z} - \mathbf{g}/L - \lambda_1/L]_+$ (non-neg. shrinkage)
 - 7: $t^+ \leftarrow \frac{1}{2}(1 + \sqrt{1 + 4t^2})$
 - 8: $\mathbf{z} \leftarrow \alpha^+ + \frac{t-1}{t^+}(\alpha^+ - \alpha)$; $\alpha \leftarrow \alpha^+$; $t \leftarrow t^+$
 - 9: end for
 - 10: repeat lines 2–9 with $\mathbf{D}^{(2)}$ on input α to obtain $\alpha^{(2)}$
 - 11: return $\alpha^{(2)}$
-

remains the center frame’s mask; the first dictionary $\mathbf{D}^{(1)} \in \mathbb{R}^{(2c+1)F \times n_u}$ absorbs the wider input and all later dimensions are unchanged. The dictionaries $\mathbf{D}^{(\ell)}$ and shrinkage parameters λ_1, λ_2 are trained by back-propagation through the unrolled iterations, minimizing a masked-magnitude reconstruction loss against the ideal ratio mask with an explicit sparsity penalty on the codes,

$$\mathcal{L} = \|(\mathbf{M}_v - \mathbf{M}_v^{\text{IRM}}) \odot \mathbf{M}\|_1 + \beta \sum_{\ell} \|\alpha^{(\ell)}\|_1. \quad (13)$$

Algorithm 1 gives the per-module computation. Equations (10) to (12) make explicit that the network is a learned sparse-recovery operator: Equation (12) is one unrolled iteration of the same proximal solver used for ℓ_1 recovery, with $\mathbf{D}$ promoted to a trainable parameter, exactly the LISTA construction (Gregor & LeCun, 2010) specialized to non-negative codes and a masking output. The clamp $[\cdot]_+$ is the proximal operator of the indicator of the nonnegative orthant composed with the soft-threshold, i.e. the prox of $\lambda_1 \|\alpha\|_1 + \iota_{\alpha \geq 0}(\alpha)$, so each layer is a genuine proximal-gradient step. Run to convergence rather than truncated, the accelerated iteration attains the optimal first-order rate $F(\alpha_k) - F(\alpha^*) = O(1/k^2)$ (Beck & Teboulle, 2009); unrolling a fixed K trades this asymptotic guarantee for a learned operator that reaches comparable quality in a handful of layers, which is why inference is a single fast forward pass rather than an iterative solve (Section 6).

5.5. Joint sparsity for stereo

Stereo channels share active sources, so their codes share a row support. Stacking the per-frame channel

magnitudes into $\mathbf{Y} = [\mathbf{y}_L \ \mathbf{y}_R] \in \mathbb{R}^{F \times 2}$, we apply simultaneous OMP (Tropp et al., 2006), which scores atoms by the row norm $\|(\mathbf{D}^\top \mathbf{R})_{n,:}\|_2$ of the residual and so forces both channels to agree on the active atoms. The mixed norm $\|\mathbf{X}\|_{1,2} = \sum_n \|\mathbf{X}_{n,:}\|_2$ is the convex surrogate, and the shared support strictly improves identifiability: a row- s -sparse $\mathbf{X}$ is unique once $s < (\text{spark}(\mathbf{A}) - 1 + \text{rank}(\mathbf{Y}))/2$.

6. Experiments

Data and protocol. We use the MUSDB18 benchmark (Rafii et al., 2017), separating each mixture into vocals and accompaniment. Dictionaries and the deep model are trained on the training split; all numbers are reported on 25 held-out test tracks at 22.05 kHz with a 2048-point STFT and hop 512. We report the median scale-invariant SDR (Le Roux et al., 2019) and SDR (Vincent et al., 2006) per source. Two oracle masks bound any masking method: the ideal ratio mask (IRM) and ideal binary mask (IBM).

Methods. We compare two training-free baselines, median-filtering harmonic/percussive separation (HPSS) (FitzGerald, 2010) and REPET-SIM (Rafii & Pardo, 2013), against the four structured-recovery separators of Section 5: RPCA, supervised NMF ($K=64$ atoms per source, KL divergence), discriminative K-SVD ($K=96$, sparsity $s=10$), and the deep sparse coding network (SCN; four composite modules, $n_u=224$, $n_d=112$, $K=16$ unrolled FISTA steps, a ± 2 -frame temporal context window, 1.44M parameters, trained for 180 epochs on Apple MPS).

Main results. Table 2 and Figure 7 show a consistent ordering. The trivial floors calibrate the scale: returning the mixture unchanged already scores +4.0 dB on accompaniment, because the accompaniment dominates the mixture energy, so accompaniment SI-SDR must be read against this floor rather than against zero; a random mask is near the no-information floor on both sources. The training-free baselines are mixed: HPSS over-suppresses tonal vocal energy (its median vocal SI-SDR falls below even the mixture floor), and REPET-SIM helps only when the background strictly repeats. The supervised dictionary methods improve sharply over all floors, and the deep sparse coding network is the best non-oracle separator on both sources (2.0 dB vocals, 8.3 dB accompaniment) while also being the fastest learned model, because its forward pass is a fixed number of matrix multiplies rather than an iterative solve. A geometric-mean fusion of the three learned masks ties the deep model on vocals (2.1 dB) but no longer improves accompaniment once the deep

Table 2. Median SI-SDR and SDR (dB) on 25 MUSDB18 test tracks, with mean inference time per 6 s clip. Trivial floors, training-free baselines, structured-recovery methods, and oracle ceilings. Best non-oracle in bold.

Method	SI-SDR		SDR	Time
	Vocals	Accomp.	Vocals	(s)
Trivial floors				
Mixture (no split)	−3.7	4.0	—	0.00
Random mask	−3.9	2.9	—	0.01
Low-rank (rank 4)	−0.1	6.1	—	0.06
Training-free baselines				
HPSS (median)	−11.9	−1.4	0.3	0.25
REPET-SIM	−0.7	2.9	2.5	0.08
Structured-recovery methods				
RPCA	−3.1	0.2	−1.3	1.14
NMF (supervised)	0.5	5.1	1.8	0.46
K-SVD + SRC	0.8	7.1	2.7	0.28
SCN (ours)	2.0	8.3	4.1	0.12
Mask fusion	2.1	8.1	—	0.85
Oracle ceilings				
Oracle IRM	10.1	15.3	10.2	—
Oracle IBM	10.2	15.8	10.5	—

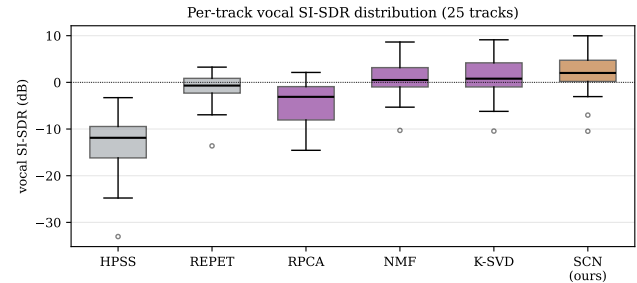


Figure 4. Per-track vocal SI-SDR over the 25 test tracks. Boxes span the interquartile range; the deep model shifts the full distribution, not only the median.

model is this strong, so we report the single model as the headline. All methods remain below the IRM/IBM oracle ceilings, as any mixture-phase magnitude mask must.

Per-track variance. Medians hide spread. Figure 4 plots the per-track vocal SI-SDR distribution. The structured-recovery methods are not uniformly better track by track: their boxes overlap, and every method has a long lower tail on dense, vocal-quiet mixtures where the sparsity prior is weakest. The deep model has the highest median and upper quartile, and shifts the whole distribution rightward relative to the classical methods rather than only improving easy tracks.

Soft versus binary masks. The masking strategy matters independently of the model. Figure 5 compares

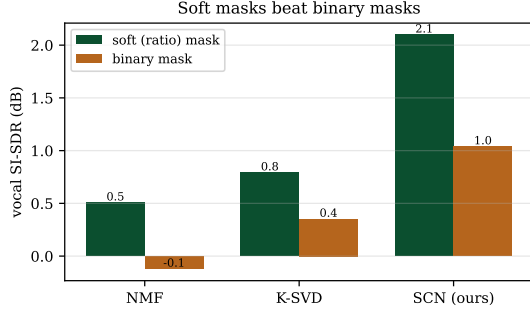


Figure 5. Soft (ratio) masks beat hard binary masks for every learned method, by up to 1 dB of vocal SI-SDR.

Table 3. Model size and per-clip cost. Trainable parameters count learned dictionaries and shrinkage parameters; classical methods store fixed dictionaries.

Method	Train?	Params	Test (s)
RPCA	no	—	1.14
NMF	dict. only	0.13M	0.46
K-SVD	dict. only	0.20M	0.28
SCN (ours)	end-to-end	1.44M	0.12

each learned method’s soft (ratio) mask against its hard-thresholded binary version. Soft masks are uniformly better, by roughly 1 dB for the deep model (2.1 vs 1.0 dB), because a continuous gain preserves the partial-overlap cells that a binary decision discards. This mirrors the gap between the IRM and IBM oracles and motivates our soft-mask training target.

Stereo joint sparsity. Evaluating the simultaneous-OMP stereo separator of Section 5.5 on 10 stereo test tracks gives median SI-SDR 0.9 dB vocals and 7.3 dB accompaniment, matching the mono K-SVD dictionary it is built on while producing a coherent stereo image, since the shared row support forces the two channels to agree on active atoms. This confirms the joint-sparsity extension is sound; a larger stereo study is left to future work.

Cost and the oracle gap. Table 3 contrasts model size and per-clip cost. The deep coder is the smallest learned model that we train and, because inference is a fixed number of matrix multiplies, the fastest at test time, an order of magnitude quicker than the iterative RPCA solve. Figure 6 reframes the scores as the fraction of the IRM oracle each method reaches on accompaniment: the supervised methods recover roughly half of the oracle, the deep coder a little more, and a wide gap to the ceiling remains, set largely by mixture-phase magnitude masking.

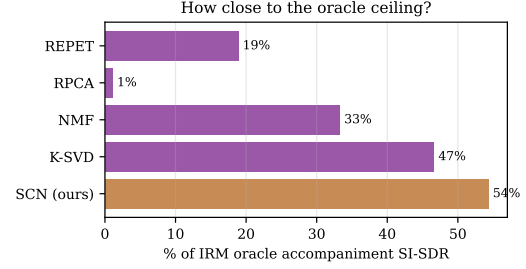


Figure 6. Fraction of the ideal-ratio-mask oracle accompaniment SI-SDR reached by each method. A large headroom to the ceiling remains, dominated by the mixture-phase magnitude-masking assumption.

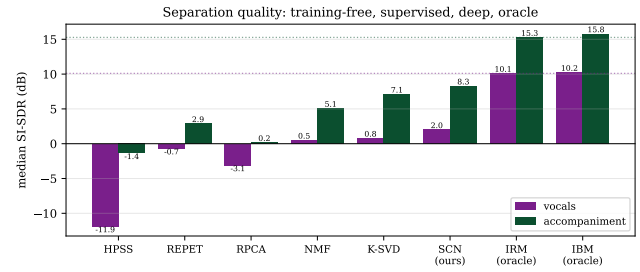


Figure 7. Median SI-SDR by method. Dotted lines mark the oracle ceilings. Supervision and unrolled learning each add a clear increment over the training-free baselines.

Spectrogram structure. Figure 8 shows the sparsity the methods exploit: the vocal magnitude concentrates on a thin harmonic scaffold while the accompaniment forms a denser, lower-rank stack. Figure 9 reports the RPCA decomposition and confirms the low-rank prior: the magnitude spectrogram’s singular spectrum decays sharply and the proximal iteration converges to a relative residual below 10^{-7} .

Ablations. Table 4 sweeps the central hyperparameters and the trends are informative rather than uniform. For supervised NMF the vocal SI-SDR is highest at the smallest dictionary ($K=16$) and declines as K grows: with limited training data a small dictionary acts as a stronger prior and generalizes better, a regularization effect rather than a capacity limit. K-SVD separation improves with the per-frame sparsity budget up to $s=10$ and then saturates, so richer codes help up to the point where added atoms stop explaining new source-specific structure. For RPCA the vocal SI-SDR increases monotonically with the penalty λ : a larger penalty pushes more energy into the low-rank accompaniment and leaves a cleaner sparse vocal residual, so the canonical $\lambda=1/\sqrt{\max(F, T)}$ is conservative for the singing-voice task even though it is the value certified for exact low-rank/sparse recovery.

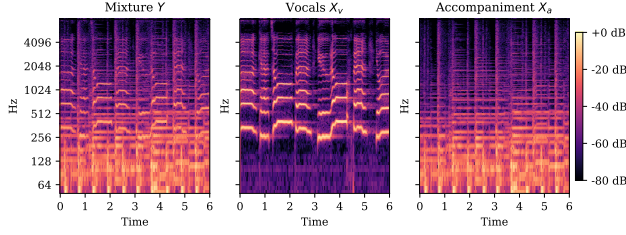


Figure 8. Log-magnitude STFT of a test clip. Vocals are sparse on a formant scaffold; accompaniment is denser and lower-rank.

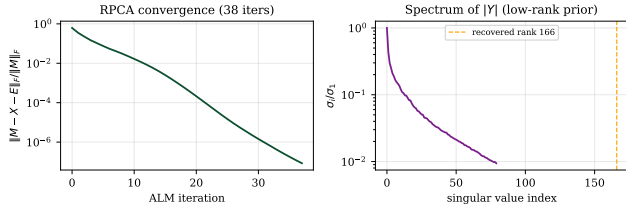


Figure 9. Left: relative residual of the principal-component-pursuit iteration. Right: normalized singular spectrum of the mixture magnitude, justifying the low-rank accompaniment prior.

Learned dictionaries. Figure 10 shows the supervised NMF atoms sorted by spectral centroid: vocal atoms concentrate harmonic ridges in the formant region while accompaniment atoms spread lower and wider, the parts-based structure NMF is designed to recover. Figure 11 shows that the deep model trains stably and keeps its intermediate codes sparse, confirming the sparsity prior is active rather than decorative.

6.1. Coherence: a negative result

The recovery condition Equation (4) requires low coherence, but learned overcomplete dictionaries are built for reconstruction, not incoherence. Figure 12 reports the off-diagonal Gram histogram of the K-SVD vocal dictionary: the coherence is $\mu \approx 0.85$, so Equation (4) certifies exact recovery only for $s < \frac{1}{2}(1 + 1/\mu) \approx 1.1$, far below the $s=10$ at which the method operates well. The worst-case guarantee is therefore vacuous in this regime, yet separation succeeds. This is the expected average-case-versus-worst-case gap: coherence bounds the worst sparse vector, while typical music frames are far from that adversarial case. We report it because it sets honest expectations for how much of the classical theory transfers to trained audio dictionaries.

Table 4. Ablations (median vocal SI-SDR, dB, 12 tracks).

NMF atoms K	16	32	64	128
Vocal SI-SDR	2.2	1.9	1.8	1.5
K-SVD sparsity s	3	5	10	15
Vocal SI-SDR	1.4	2.3	2.7	2.7
RPCA λ scale	0.5	1.0	1.5	2.0
Vocal SI-SDR	-3.9	-3.6	-3.2	-2.3

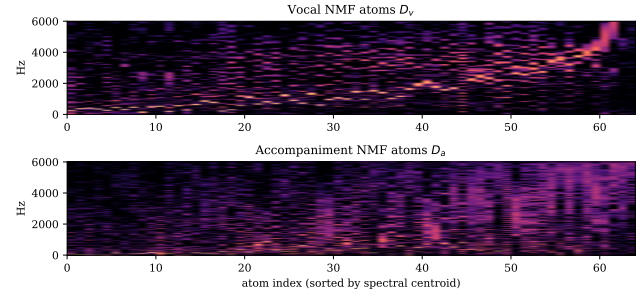


Figure 10. Supervised NMF dictionaries, atoms sorted by spectral centroid. Vocal atoms (top) versus accompaniment atoms (bottom).

7. Discussion

The results are deliberately scoped: short clips, modest dictionaries, and a 1.4M-parameter network trained on a laptop GPU. Large end-to-end systems reach substantially higher vocal SI-SDR with three to four orders of magnitude more parameters and data (Rouard et al., 2023; Mitsufuji et al., 2022). The value here is the unifying view and the controlled comparison: on one front end, the same axis of increasing structure-learning, from training-free decomposition, to supervised factorization, to discriminative dictionaries, to an unrolled deep solver, produces a monotone improvement, and the unrolled solver is both the most accurate and the cheapest to run at inference. Two limitations are intrinsic to the present setup. First, magnitude masking with mixture phase upper-bounds every method at the IRM oracle; a complex-mask or phase-reconstruction extension would lift the ceiling (Luo & Mesgarani, 2019). Second, the deep model uses only a short ± 2 -frame context; the gain it brings suggests fully convolutional or recurrent composite modules with a wider receptive field as a promising next step. The stereo joint-sparsity extension of Section 5.5 also remains to be evaluated at scale.

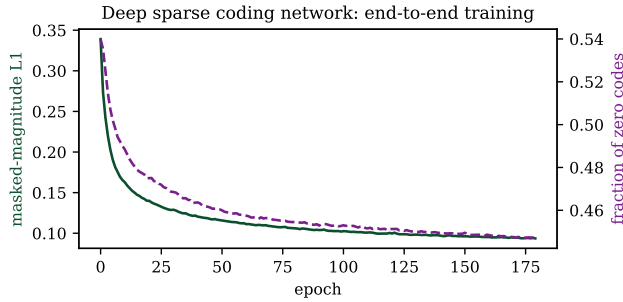


Figure 11. Deep sparse coding network training. Masked-magnitude reconstruction loss (left axis) and fraction of zero codes (right axis) across epochs.

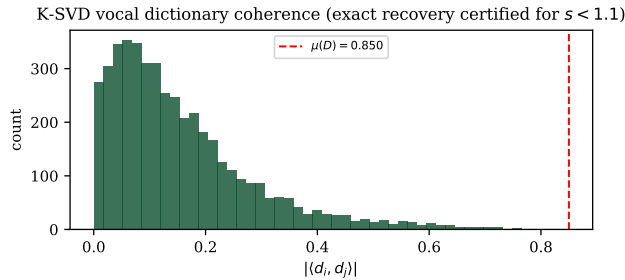


Figure 12. Off-diagonal coherence of the learned K-SVD vocal dictionary. The worst-case recovery bound certifies only $s \lesssim 1$, but separation at $s=10$ works, evidence of an average-case-versus-worst-case gap.

8. Conclusion

We presented a unified compressed-sensing view of music source separation, casting low-rank decomposition, supervised NMF, discriminative dictionary learning, and a deep unrolled sparse coder as points on one structure-recovery spectrum over the STFT. On MUSDB18 the methods order cleanly, the unrolled deep sparse coding network is the strongest non-oracle separator while being the fastest learned model, and a coherence analysis delimits where the classical recovery theory does and does not transfer to learned audio dictionaries. Code, models, and audio are released.

Reproducibility

All code, trained dictionaries and network weights, the evaluation harness, and curated before/after audio are available at https://github.com/takakhoo/ENG109_Final_Project.

References

Aharon, M., Elad, M., and Bruckstein, A. K-SVD: An algorithm for designing overcomplete dictionary

for sparse representation. *IEEE Trans. Signal Processing*, 54(11):4311–4322, 2006.

Araki, S., Ito, N., Haeb-Umbach, R., Wichern, G., Wang, Z.-Q., and Mitsufuji, Y. 30+ years of source separation research: Achievements and future challenges. In *IEEE Int. Conf. Acoustics, Speech and Signal Processing (ICASSP)*, 2025.

Beck, A. and Teboulle, M. A fast iterative shrinkage-thresholding algorithm for linear inverse problems. *SIAM Journal on Imaging Sciences*, 2(1):183–202, 2009.

Candès, E. J. and Recht, B. Exact matrix completion via convex optimization. *Foundations of Computational Mathematics*, 9(6):717–772, 2009.

Candès, E. J. and Tao, T. Decoding by linear programming. *IEEE Trans. Information Theory*, 51(12):4203–4215, 2005.

Candès, E. J., Li, X., Ma, Y., and Wright, J. Robust principal component analysis? *Journal of the ACM*, 58(3):1–37, 2011.

Cano, E., FitzGerald, D., Liutkus, A., Plumbley, M. D., and Stöter, F.-R. Musical source separation: An introduction. *IEEE Signal Processing Magazine*, 36(1):31–40, 2019.

Défossez, A., Usunier, N., Bottou, L., and Bach, F. Music source separation in the waveform domain. *arXiv:1911.13254*, 2021.

Donoho, D. L. Compressed sensing. *IEEE Trans. Information Theory*, 52(4):1289–1306, 2006.

Elad, M. *Sparse and Redundant Representations: From Theory to Applications in Signal and Image Processing*. Springer, 2010.

Engan, K., Aase, S. O., and Husøy, J. H. Method of optimal directions for frame design. In *IEEE Int. Conf. Acoustics, Speech, and Signal Processing (ICASSP)*, pp. 2443–2446, 1999.

Févotte, C., Bertin, N., and Durrieu, J.-L. Nonnegative matrix factorization with the Itakura-Saito divergence. *Neural Computation*, 21(3):793–830, 2009.

FitzGerald, D. Harmonic/percussive separation using median filtering. In *Int. Conf. on Digital Audio Effects (DAFx)*, 2010.

Gregor, K. and LeCun, Y. Learning fast approximations of sparse coding. In *Int. Conf. on Machine Learning (ICML)*, pp. 399–406, 2010.

- Hennequin, R., Khlif, A., Voituret, F., and Moussallam, M. Spleeter: A fast and efficient music source separation tool with pre-trained models. *Journal of Open Source Software*, 5(50):2154, 2020.
- Huang, P.-S., Chen, S. D., Smaragdis, P., and Hasegawa-Johnson, M. Singing-voice separation from monaural recordings using robust principal component analysis. In *IEEE Int. Conf. Acoustics, Speech and Signal Processing (ICASSP)*, pp. 57–60, 2012.
- Jansson, A., Humphrey, E. J., Montecchio, N., Bittner, R., Kumar, A., and Weyde, T. Singing voice separation with deep U-Net convolutional networks. In *Int. Society for Music Information Retrieval (ISMIR)*, pp. 745–751, 2017.
- Le Roux, J., Hershey, J. R., and Weninger, F. Deep NMF for speech separation. In *IEEE Int. Conf. Acoustics, Speech and Signal Processing (ICASSP)*, pp. 66–70, 2015.
- Le Roux, J., Wisdom, S., Erdogan, H., and Hershey, J. R. SDR – half-baked or well done? In *IEEE Int. Conf. Acoustics, Speech and Signal Processing (ICASSP)*, pp. 626–630, 2019.
- Lee, D. D. and Seung, H. S. Learning the parts of objects by non-negative matrix factorization. *Nature*, 401(6755):788–791, 1999.
- Lee, D. D. and Seung, H. S. Algorithms for non-negative matrix factorization. In *Advances in Neural Information Processing Systems (NeurIPS)*, pp. 556–562, 2000.
- Luo, Y. and Mesgarani, N. Conv-TasNet: Surpassing ideal time-frequency magnitude masking for speech separation. *IEEE/ACM Trans. Audio, Speech, Language Process.*, 27(8):1256–1266, 2019.
- Mairal, J., Bach, F., and Ponce, J. Task-driven dictionary learning. *IEEE Trans. Pattern Analysis and Machine Intelligence*, 34(4):791–804, 2012.
- Mitsufuji, Y., Fabbro, G., Uhlich, S., Stöter, F.-R., Défossez, A., Kim, M., and Choi, W. Music demixing challenge 2021. *Frontiers in Signal Processing*, 1:808395, 2022.
- Monga, V., Li, Y., and Eldar, Y. C. Algorithm unrolling: Interpretable, efficient deep learning for signal and image processing. *IEEE Signal Processing Magazine*, 38(2):18–44, 2021.
- Papayan, V., Romano, Y., and Elad, M. Convolutional neural networks analyzed via convolutional sparse coding. *Journal of Machine Learning Research*, 18(83):1–52, 2017.
- Raffi, Z. and Pardo, B. REPET: A simple method for music/voice separation. *IEEE Trans. Audio, Speech, Language Process.*, 21(1):73–84, 2013.
- Raffi, Z., Liutkus, A., Stöter, F.-R., Mimilakis, S. I., and Bittner, R. The MUSDB18 corpus for music separation, 2017.
- Rouard, S., Massa, F., and Défossez, A. Hybrid transformers for music source separation. In *IEEE Int. Conf. Acoustics, Speech and Signal Processing (ICASSP)*, 2023.
- Smaragdis, P. Convolutional speech bases and their application to supervised speech separation. *IEEE Trans. Audio, Speech, Language Process.*, 15(1):1–12, 2007.
- Smaragdis, P. and Brown, J. C. Non-negative matrix factorization for polyphonic music transcription. In *IEEE Workshop on Applications of Signal Processing to Audio and Acoustics (WASPAA)*, pp. 177–180, 2003.
- Sprechmann, P., Bronstein, A. M., and Sapiro, G. Learning efficient sparse and low-rank models. *IEEE Trans. Pattern Analysis and Machine Intelligence*, 37(9):1821–1833, 2015.
- Stoller, D., Ewert, S., and Dixon, S. Wave-U-Net: A multi-scale neural network for end-to-end audio source separation. In *Int. Society for Music Information Retrieval (ISMIR)*, pp. 334–340, 2018.
- Stöter, F.-R., Uhlich, S., Liutkus, A., and Mitsufuji, Y. Open-unmix – a reference implementation for music source separation. *Journal of Open Source Software*, 4(41):1667, 2019.
- Sun, X., Nasrabadi, N. M., and Tran, T. D. Supervised deep sparse coding networks for image classification. *IEEE Trans. Image Processing*, 29:405–418, 2019.
- Tropp, J. A. and Gilbert, A. C. Signal recovery from random measurements via orthogonal matching pursuit. *IEEE Trans. Information Theory*, 53(12):4655–4666, 2007.
- Tropp, J. A., Gilbert, A. C., and Strauss, M. J. Algorithms for simultaneous sparse approximation. part i: Greedy pursuit. *Signal Processing*, 86(3):572–588, 2006.
- Vincent, E., Gribonval, R., and Févotte, C. Performance measurement in blind audio source separation. *IEEE Trans. Audio, Speech, Language Process.*, 14(4):1462–1469, 2006.

- Virtanen, T. Monaural sound source separation by nonnegative matrix factorization with temporal continuity and sparseness criteria. *IEEE Trans. Audio, Speech, Language Process.*, 15(3):1066–1074, 2007.
- Wright, J., Yang, A. Y., Ganesh, A., Sastry, S. S., and Ma, Y. Robust face recognition via sparse representation. *IEEE Trans. Pattern Analysis and Machine Intelligence*, 31(2):210–227, 2009.